

Errors and power

What can a study detect, miss or falsely declare?

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Type I and II errors

Power and study size

Multiple testing, p -hacking and HARKing

Does the drug have an effect?

H_0 : no effect on the outcome. H_1 : an effect on the outcome.

Decision / Reality	H_0 is true The drug has no effect	H_0 is false The drug has an effect
Reject H_0	Type I error Falsely declaring an effect	
Do not reject H_0	Correct non-rejection	

In JAMA, the outcome was disappearance of sore-throat symptoms at a specified time.

Does the drug have an effect?

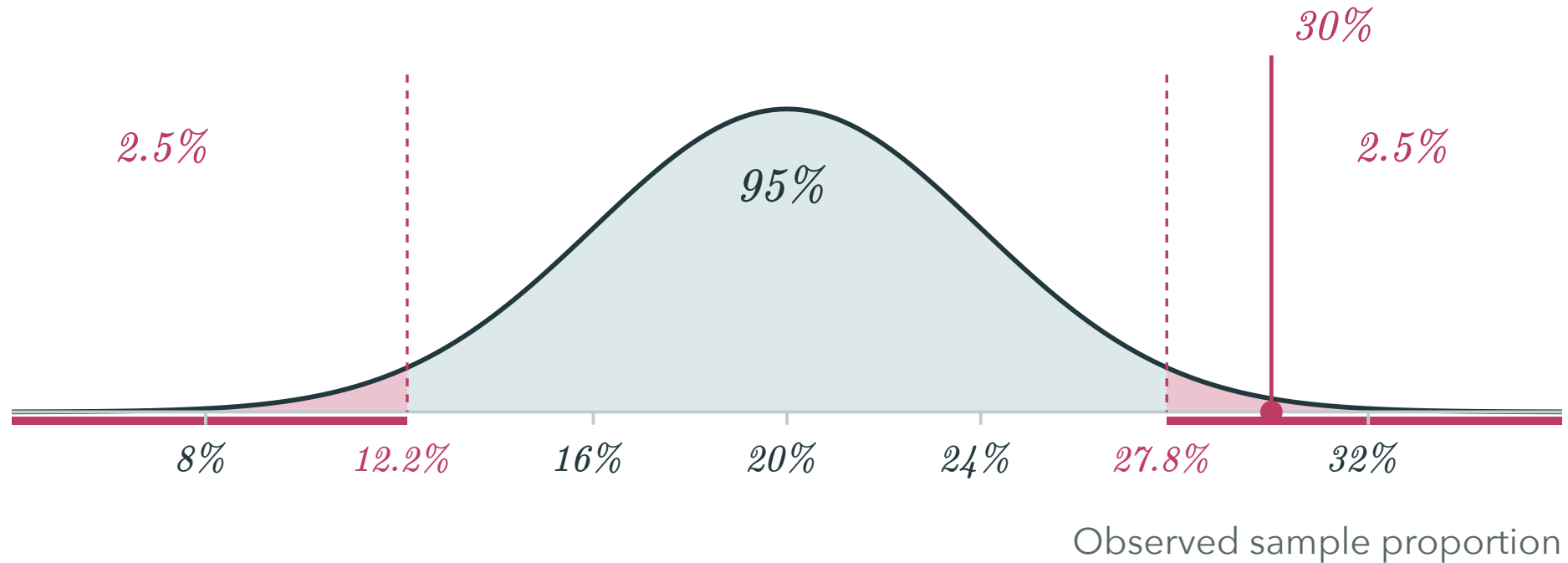
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Decision / Reality	H_0 is true The drug has no effect	H_0 is false The drug has an effect
Reject H_0	Type I error Falsely declaring an effect	Correct detection
Do not reject H_0	Correct non-rejection	Type II error Missing a real effect

Our sample gives a decision. The population determines whether it is correct.

Type I error: falsely declaring an effect

Return to the simple example: H_0 says the prevalence is 20%.



$$\alpha = \Pr(\text{reject } H_0 \mid H_0 \text{ true}) \approx 5\%$$

The 5% calibration is approximate for this proportion test.

A real difference needs a specified value

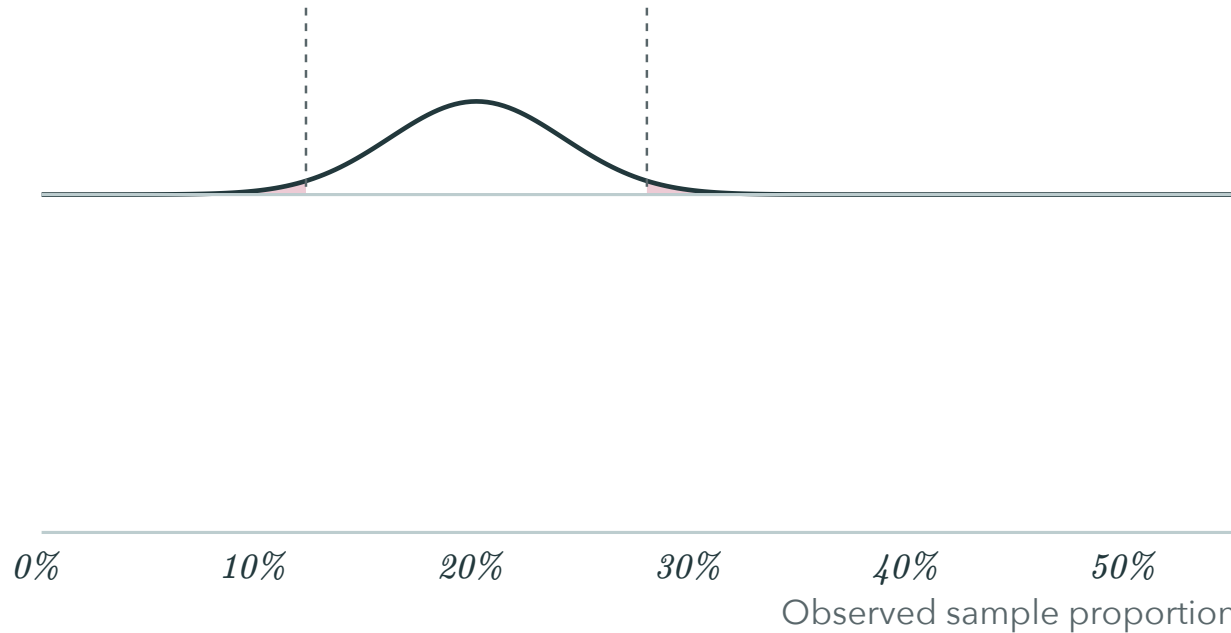
The test still asks whether prevalence is *20%*.

$$H_0 : p = 20\%$$

100 people per study

Significance level: 5%

If the true prevalence is 20%



Type I error
 $\alpha = 5\%$

What would samples look like if H_0 were false?

We specify an alternative value to calculate the chance of missing it.

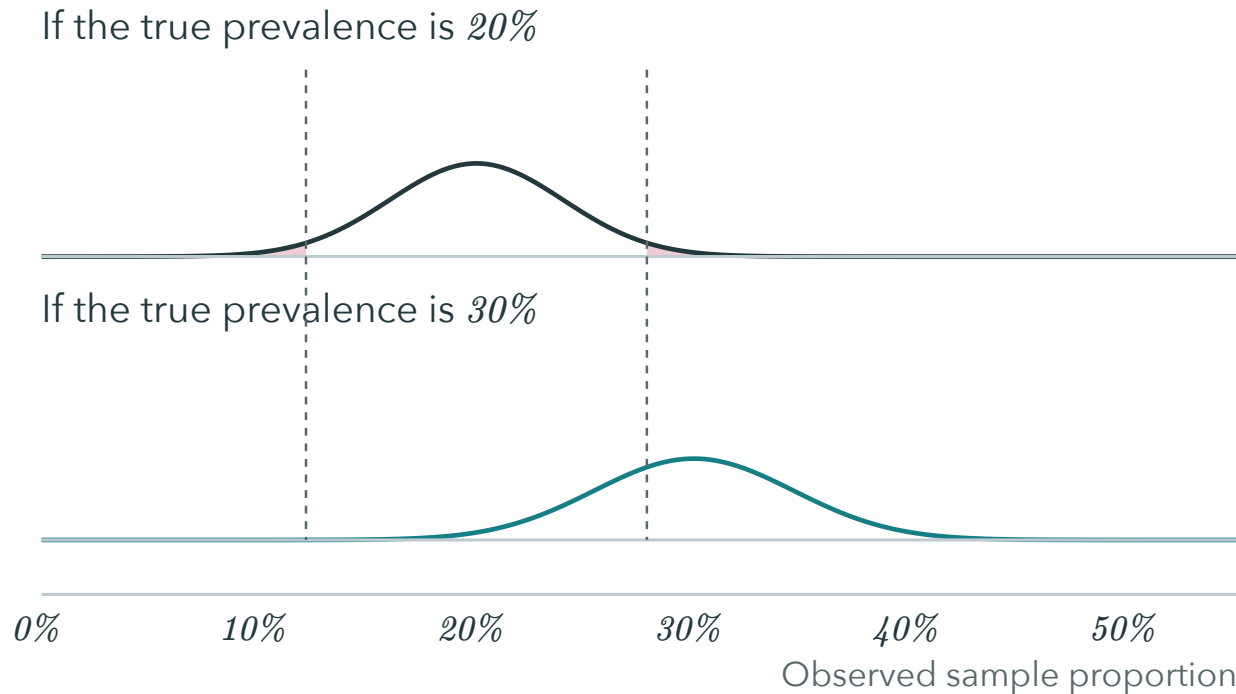
A real difference needs a specified value

The test still asks whether prevalence is *20%*.

$$H_0 : p = 20\%$$

100 people per study

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Type I error
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Suppose it is 30%. The decision boundaries still come from H_0 .

We specify an alternative value to calculate the chance of missing it.

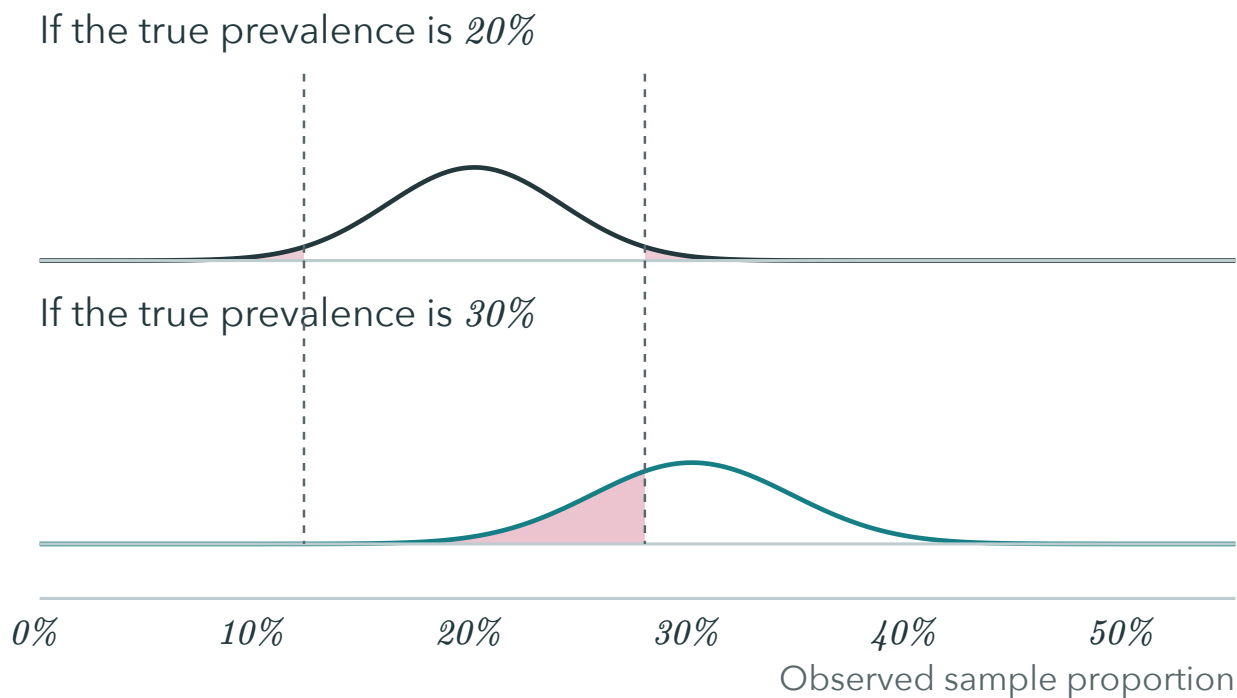
Type II error and statistical power

Suppose the true prevalence is 30%, while H_0 says 20%.

$H_0 : p = 20\%$

100 people per study

Significance level: 5%



Type I error
 $\alpha = 5\%$

Missed detection
 $\beta \approx 31.9\%$

Red area: the test does not reject, although H_0 is false.

β is the probability of missing this specified difference.

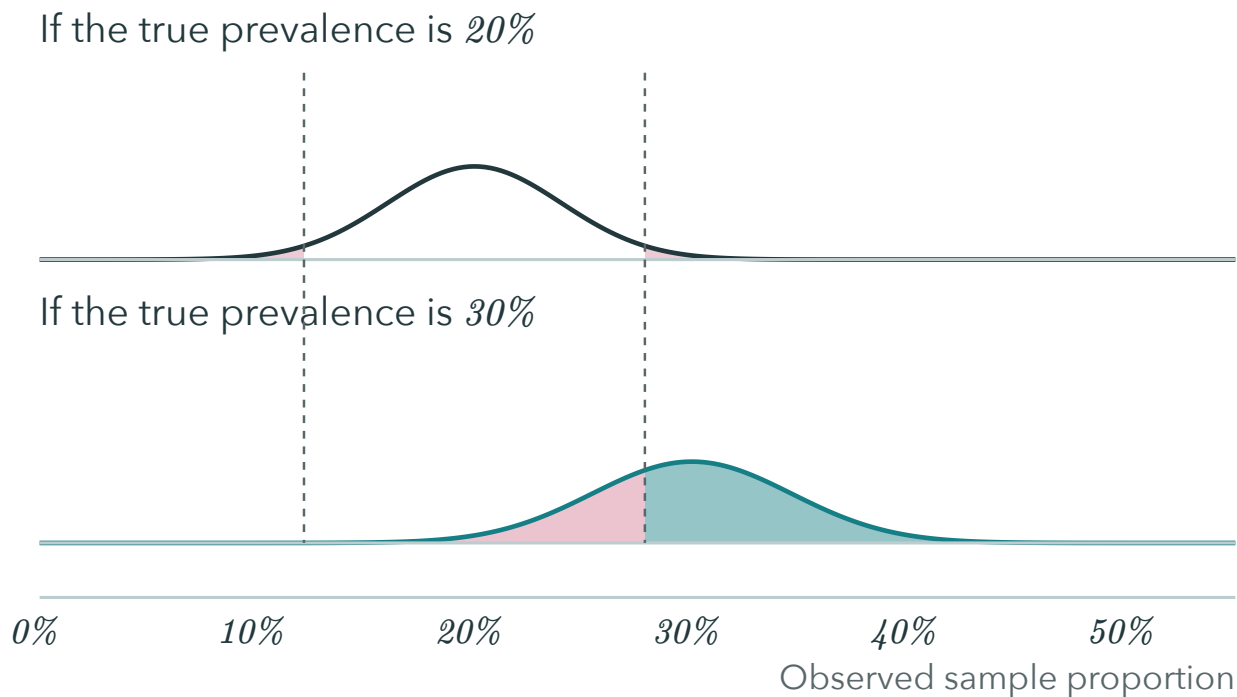
Type II error and statistical power

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Power
 68.1%

$$\text{Power} = 1 - \beta$$

Both rejection tails contribute to the green area under the alternative.

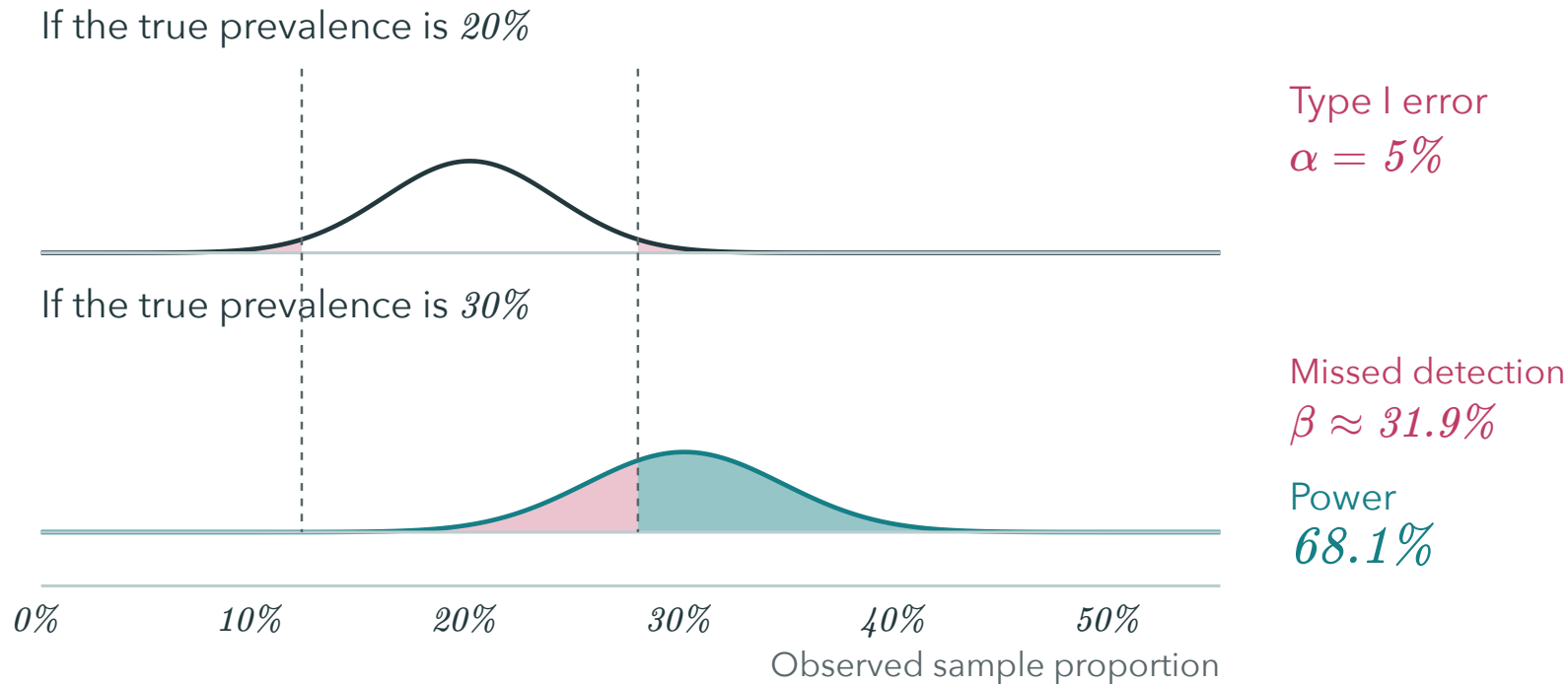
More people make estimates more precise

The true prevalence stays at 30% . The significance level stays at 5% .

$H_0 : p = 20\%$

100 people per study

Significance level: 5%



The curves narrow, the critical limits move towards 20% , and power increases.

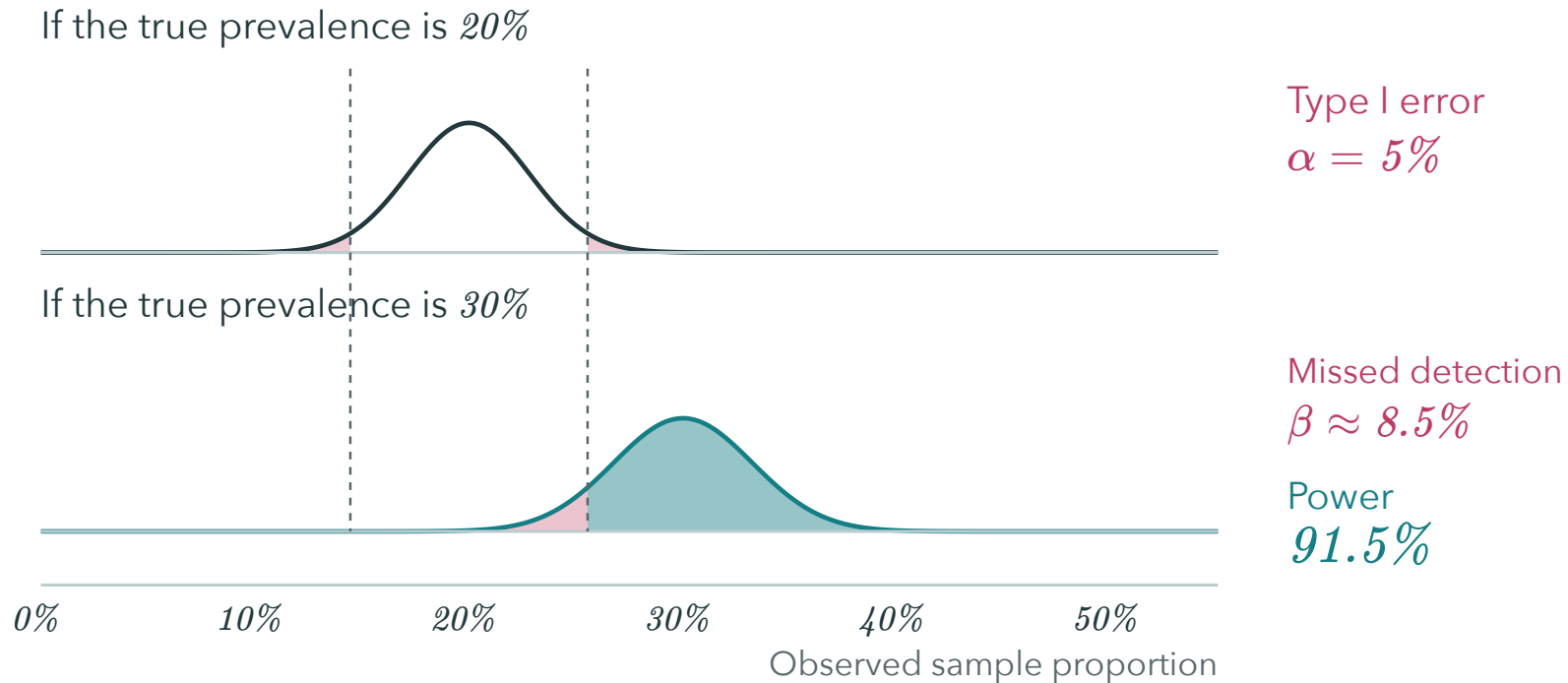
More people make estimates more precise

The true prevalence stays at 30%. The significance level stays at 5%.

$H_0 : p = 20\%$

200 people per study

Significance level: 5%



The curves narrow, the critical limits move towards 20%, and power increases.

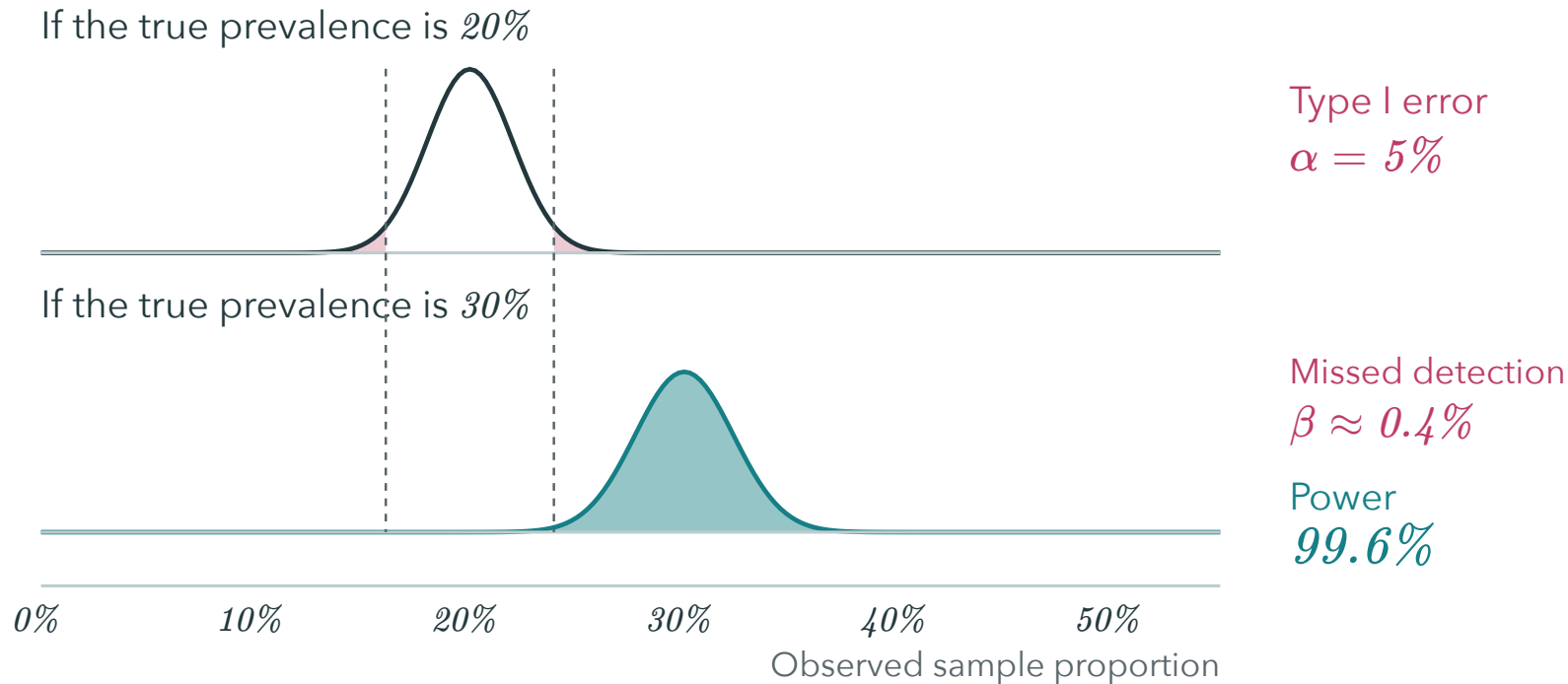
More people make estimates more precise

The true prevalence stays at 30%. The significance level stays at 5%.

$$H_0 : p = 20\%$$

400 people per study

Significance level: 5%



The curves narrow, the critical limits move towards 20%, and power increases.

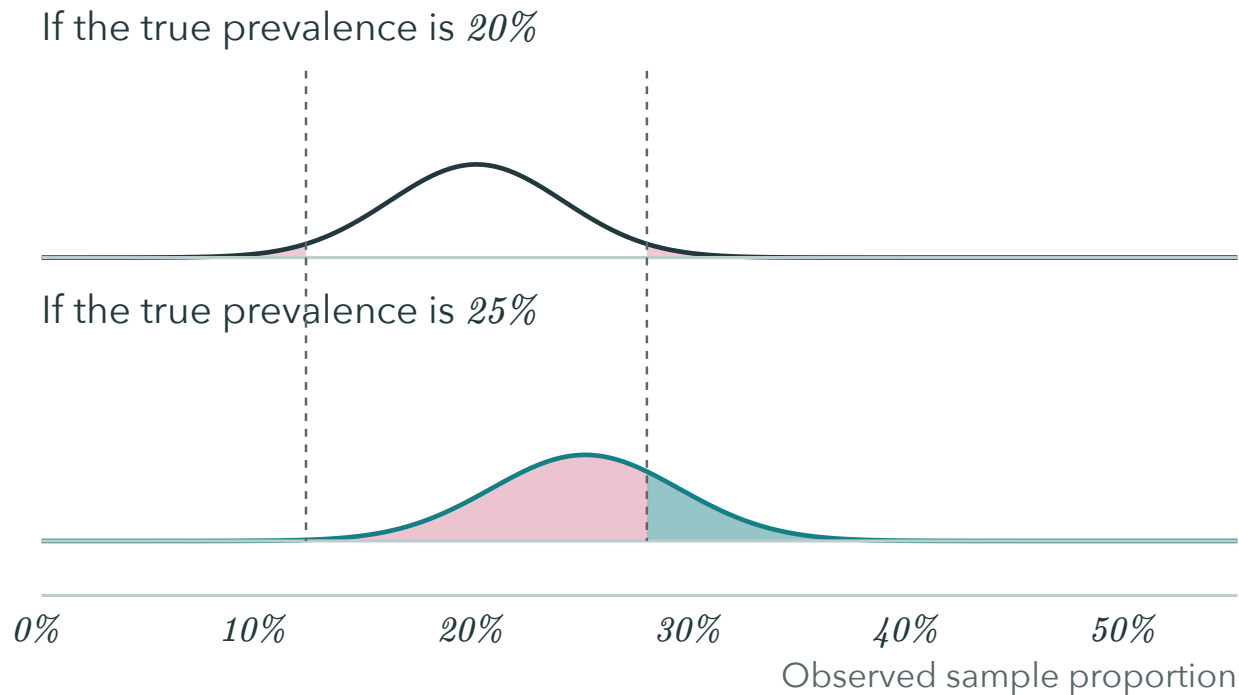
Power depends on the effect and the threshold

Keep 100 people. Consider a smaller real difference.

$$H_0 : p = 20\%$$

100 people per study

Significance level: 5%



Type I error
 $\alpha = 5\%$

Missed detection
 $\beta \approx 74.3\%$

Power
25.7%

25% rather than 20% is harder to detect.

A power statement needs a specified effect, sample size and significance level.

Power depends on the effect and the threshold

Keep 100 people. Now consider a larger real difference.

$$H_0 : p = 20\%$$

100 people per study

Significance level: 5%



Type I error
 $\alpha = 5\%$

Missed detection
 $\beta \approx 31.9\%$

Power
68.1%

30% rather than 20% is easier to detect.

A power statement needs a specified effect, sample size and significance level.

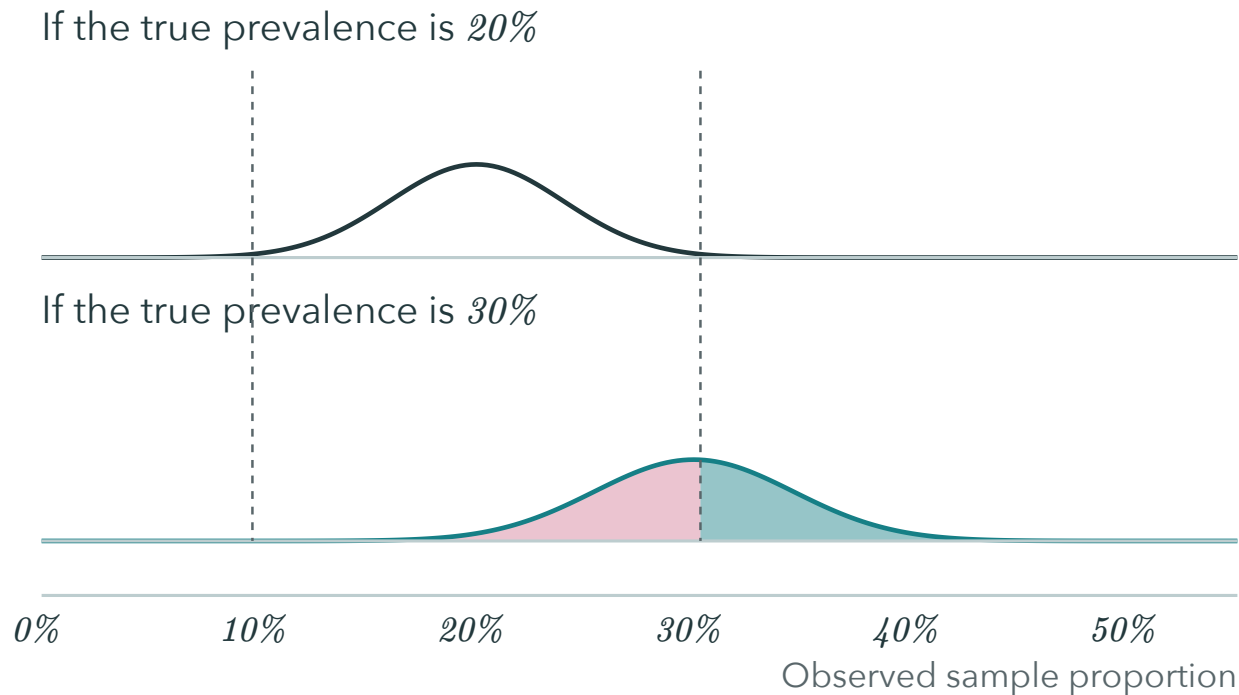
Power depends on the effect and the threshold

Keep 100 people. Now require stronger evidence before rejecting.

$$H_0 : p = 20\%$$

100 people per study

Significance level: 1%



Type I error
 $\alpha = 1\%$

Missed detection
 $\beta \approx 52.6\%$

Power
47.4%

A stricter significance level reduces the chance of detection.

A power statement needs a specified effect, sample size and significance level.

Why plan the sample size?

A real effect can exist and still be missed by an underpowered study.

Funders and research teams

Commit money and time to answer a useful question.

Participants

Contribute their time and accept burdens or risks.

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Contribute their time and accept burdens or risks.

Future patients

May wait longer for a useful treatment if its benefit is missed.

Are we giving this study a reasonable chance to answer its question?

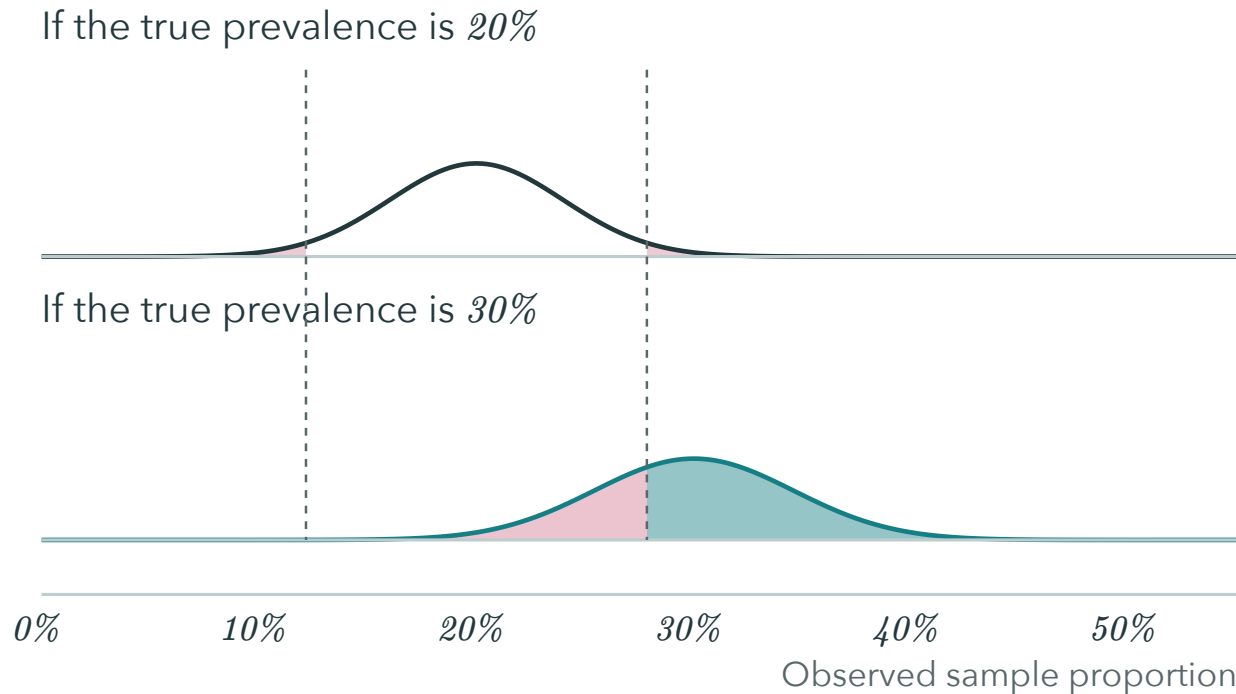
The logic of sample-size planning

If the true prevalence is *30%*, what size gives *80%* power?

$H_0 : p = 20\%$

100 people per study

Significance level: 5%



Type I error
 $\alpha = 5\%$

Missed detection
 $\beta \approx 31.9\%$

Power
68.1%

At 100 people, the detection probability is below the target.

$\alpha = 0.05$. A target power of 80% means $\beta = 0.20$ for this assumed effect.

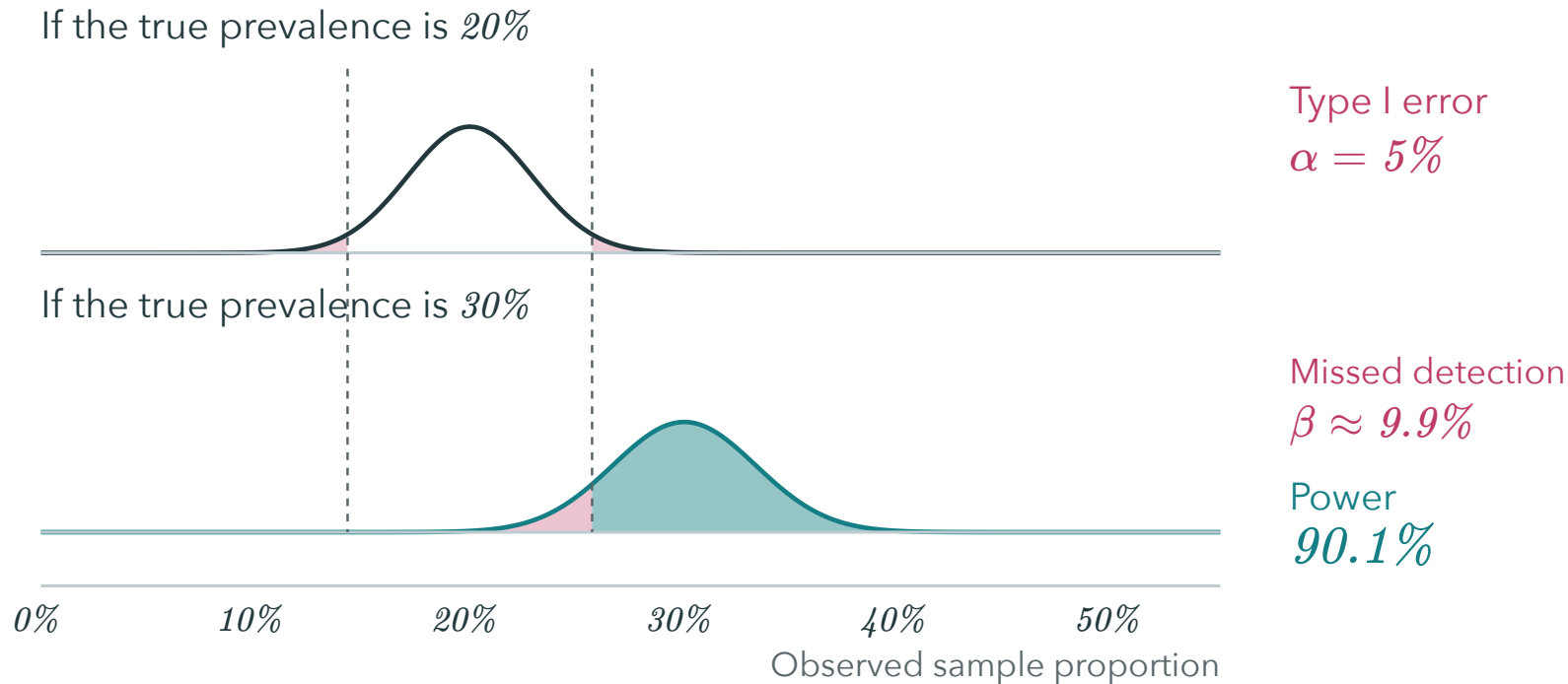
The logic of sample-size planning

If the true prevalence is *30%*, what size gives *90%* power?

$H_0 : p = 20\%$

189 people per study

Significance level: 5%



About 189 people reach 90% power in this Normal approximation.

$\alpha = 0.05$. A target power of 90% means $\beta = 0.10$ for this assumed effect.

What the JAMA study planned

Two treatment groups. Primary outcome: symptom resolution at 24 hours.

Placebo assumption

11%

Dexamethasone assumption

29%

Under the null hypothesis, the two groups have the same resolution rate.

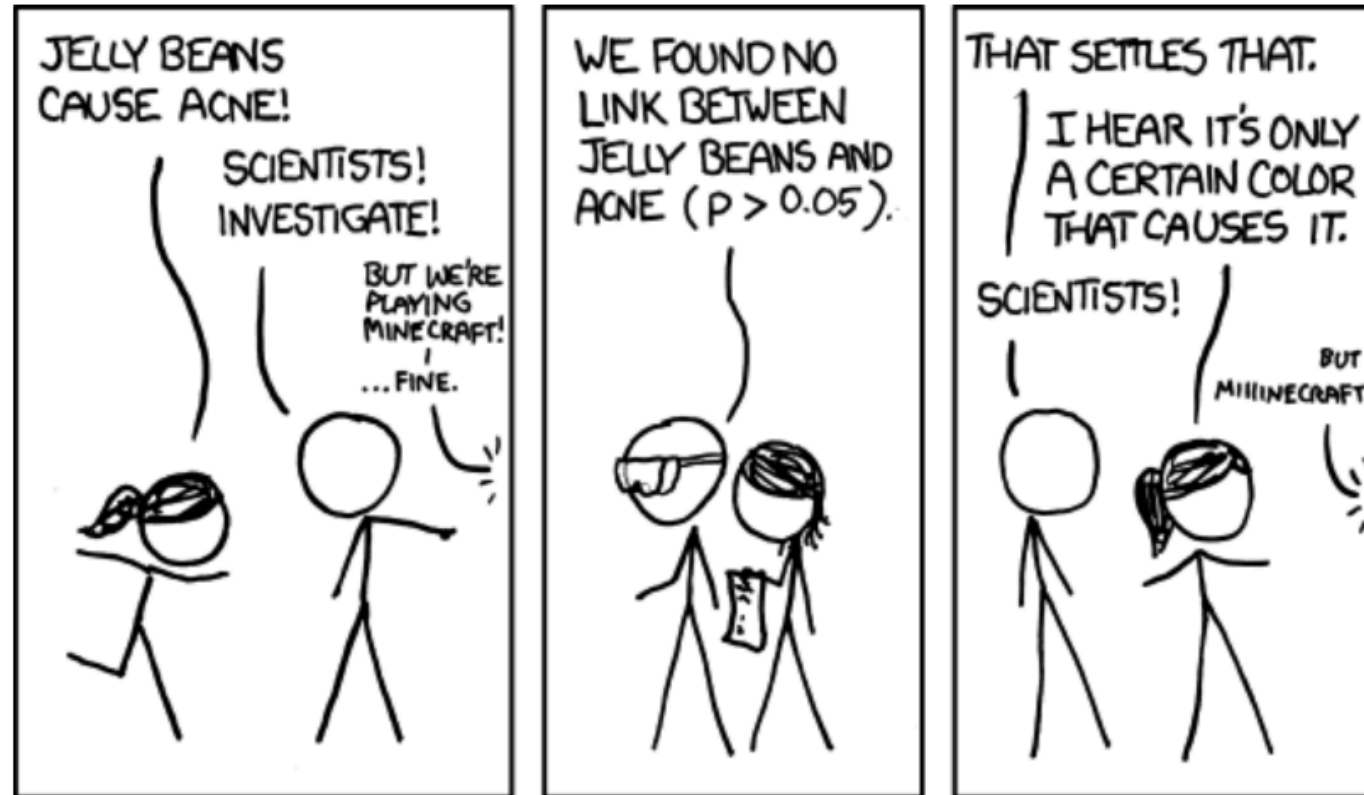
Planned alternative: $29\% - 11\% = 18\%$ absolute difference.

Significance level: 5% . Target power: 90% .

The article used two-sided p -values: a difference in either direction.

Jelly beans and acne

One planned question: are jelly beans linked to acne?

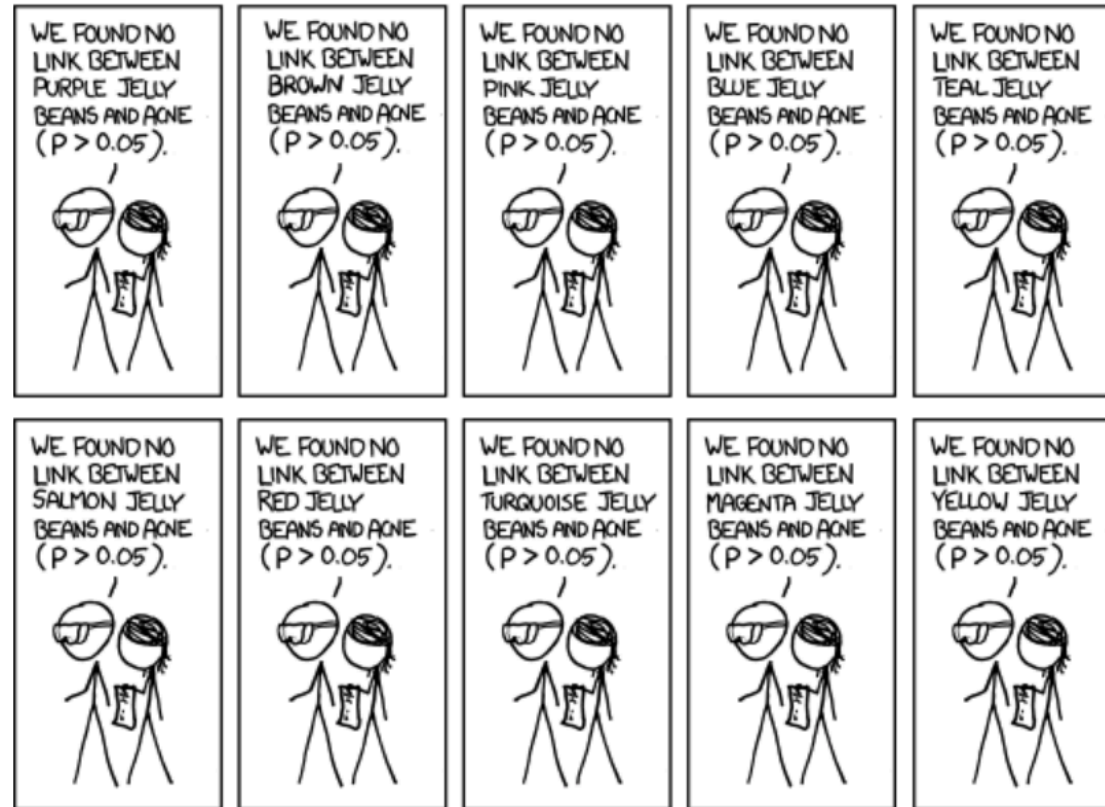


xkcd: Significant, Randall Munroe · xkcd.com/882

What happens to a 5% risk when one question becomes many?

Jelly beans and acne

Now test the colours one by one.

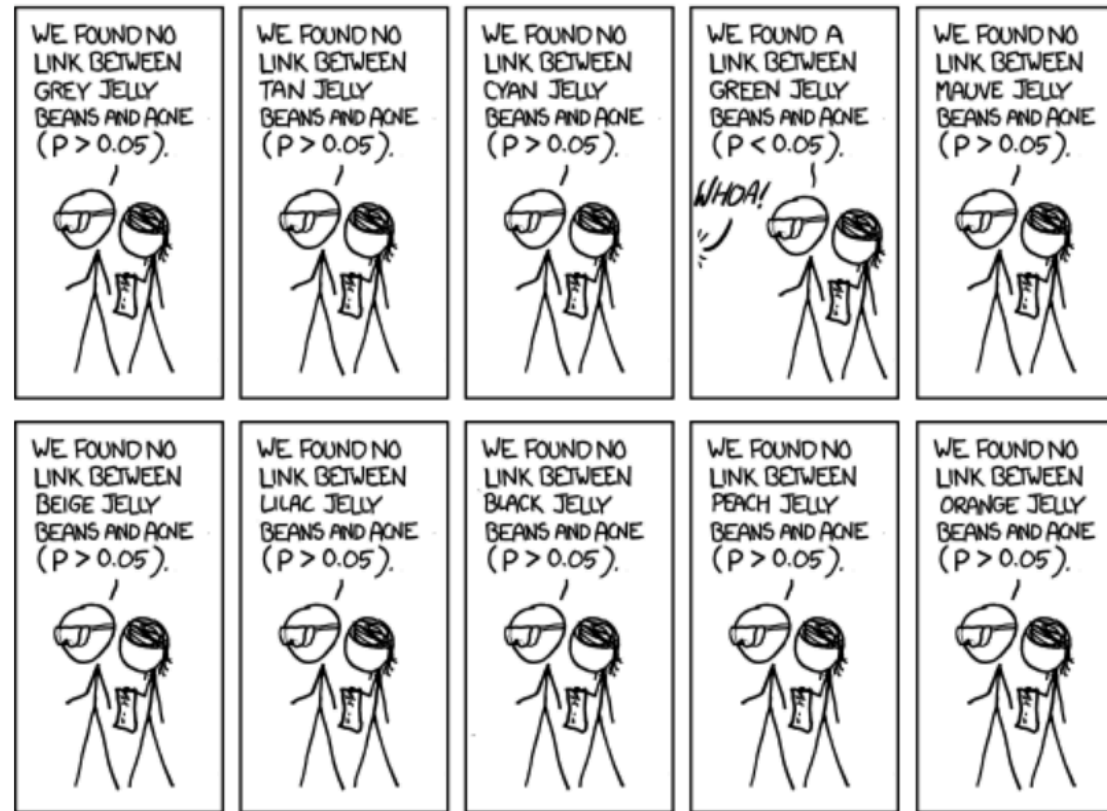


xkcd: Significant, Randall Munroe · xkcd.com/882

The reader needs to know how many questions the researchers examined.

Jelly beans and acne

Twenty colours create twenty opportunities to find a result.

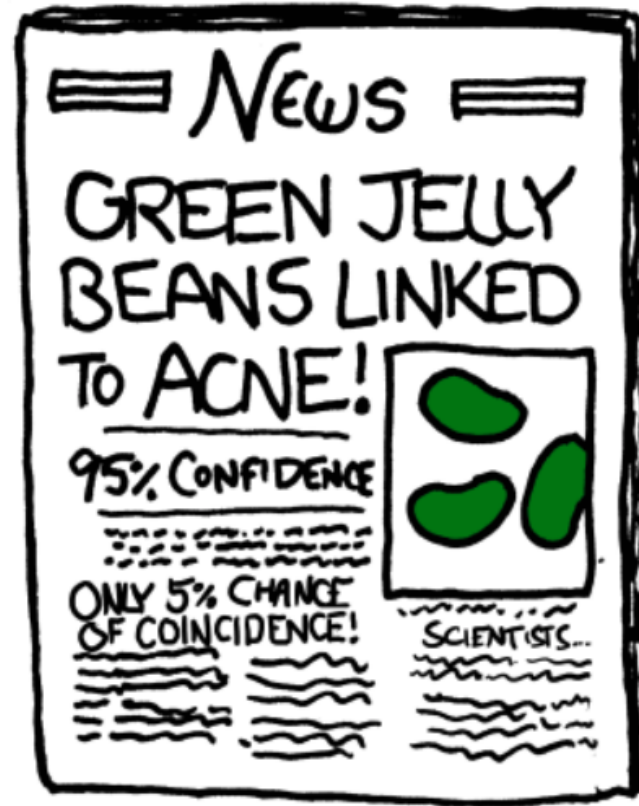


xkcd: Significant, Randall Munroe · xkcd.com/882

The reader needs to know how many questions the researchers examined.

Jelly beans and acne

The headline reports only the green result.



xkcd: Significant, Randall Munroe · xkcd.com/882

The reader needs to know how many questions the researchers examined.

A 5% risk for each test

Imagine 20 independent tests, with all 20 null hypotheses true.

Each test has a 5% chance of a false rejection.

$$\Pr(\text{no false rejection}) = 0.95^{20} \approx 0.36$$

$$\Pr(\text{at least one false rejection}) = 1 - 0.95^{20} \approx 0.64$$

About 64% of such sets contain at least one false rejection.

P-hacking

P-hacking refers to using flexible analyses or repeated testing to search for a statistically significant result, then reporting only the result that worked.

The reported result does not show the full search.

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Purple <i>p</i> = 0.54	Brown <i>p</i> = 0.82	Pink <i>p</i> = 0.13	Blue <i>p</i> = 0.61	Teal <i>p</i> = 0.34	Salmon <i>p</i> = 0.72	Red <i>p</i> = 0.19	Turquoise <i>p</i> = 0.93	Magenta <i>p</i> = 0.41	Yellow <i>p</i> = 0.27
Grey <i>p</i> = 0.63	Tan <i>p</i> = 0.16	Cyan <i>p</i> = 0.88	Mauve <i>p</i> = 0.37	Green <i>p</i> = 0.02	Peach <i>p</i> = 0.46	Orange <i>p</i> = 0.57	Black <i>p</i> = 0.31	Beige <i>p</i> = 0.68	Lilac <i>p</i> = 0.24

Illustrative significance values, not data from the cartoon.

Twenty colours create twenty opportunities to obtain a significant result.

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Illustrative significance values, not data from the cartoon.

Reporting only green hides the other nineteen tests.

Bonferroni for a planned family of tests

Aim for at most a 5% chance of any false rejection across the family.

$$\text{significance level for each test} = \frac{\alpha}{m}$$

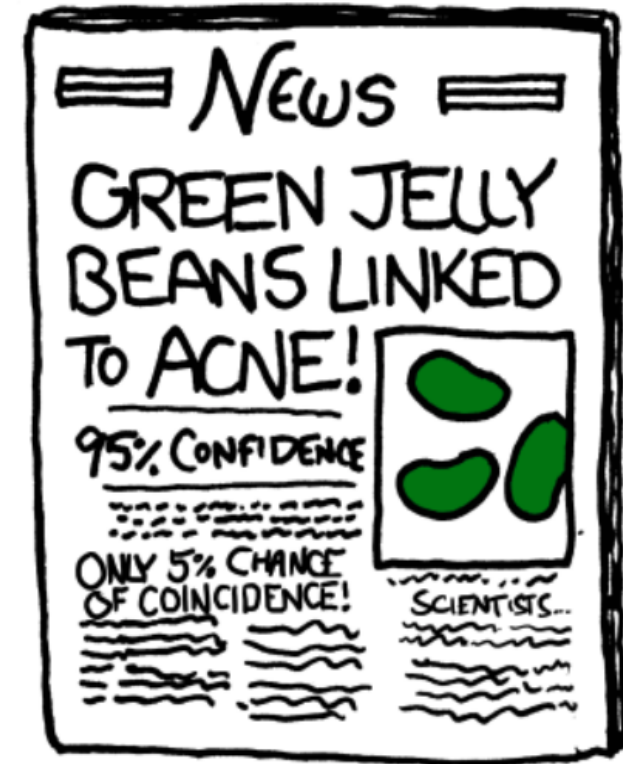
$$m = 20, \quad \alpha = 0.05 : \quad \frac{0.05}{20} = 0.0025$$

| A p -value of 0.02 would no longer pass this threshold.

Bonferroni addresses a defined set of tests, not a hidden adaptive search.

HARKing

HARKing means Hypothesizing After the Results are Known.



Source: xkcd, "Significant," Randall Munroe · xkcd.com/882

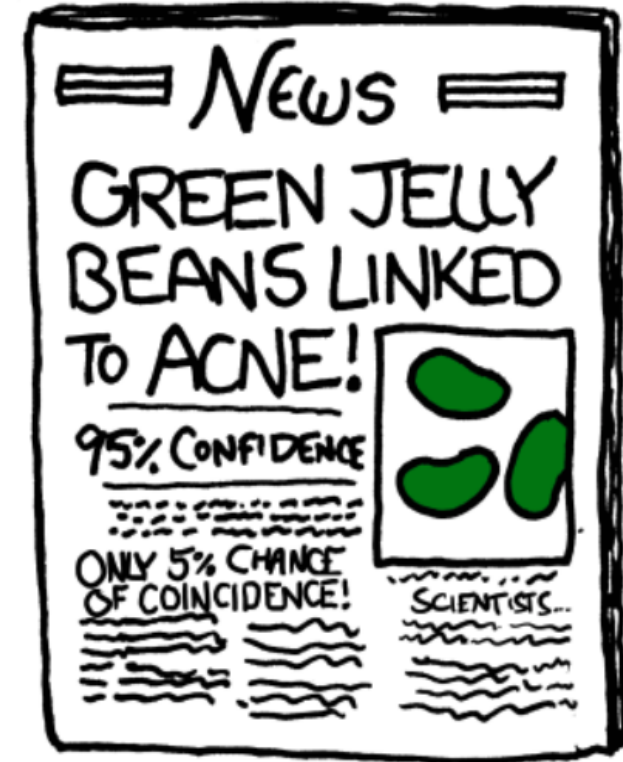
New hypotheses are useful. Their exploratory origin should be clear.

HARKing

HARKing means Hypothesizing After the Results are Known.

A **post hoc** hypothesis is presented as if it were an **a priori** hypothesis.

1. Test the jelly-bean colours.
2. Green happens to be significant.
3. Present green as the original hypothesis.

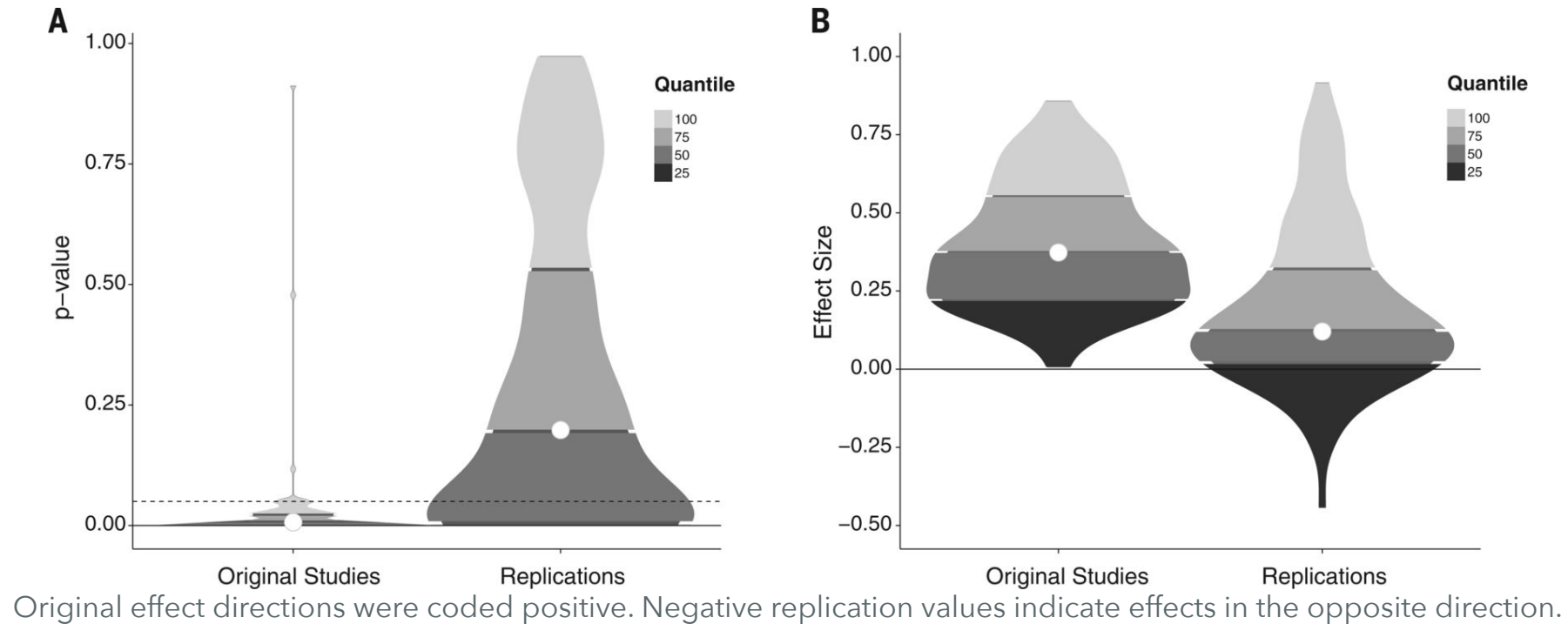


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New hypotheses are useful. Their exploratory origin should be clear.

Replicability of published psychology findings

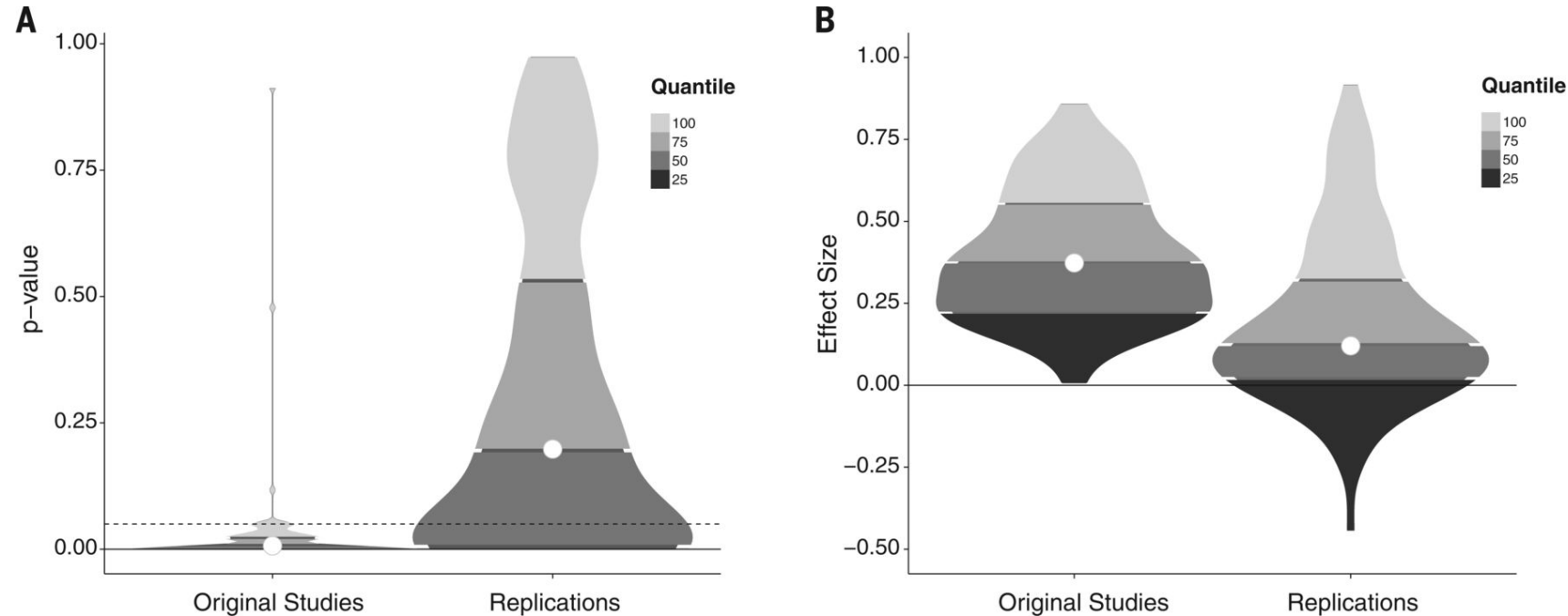
The Open Science Collaboration repeated *100* published psychology studies.



Open Science Collaboration, *Science*, 2015. Figure 1. doi:10.1126/science.aac4716

Replicability of published psychology findings

The Open Science Collaboration repeated *100* published psychology studies.



Original effect directions were coded positive. Negative replication values indicate effects in the opposite direction.

Significant results

97% in the original studies; 36% in the replications.

Mean effect size

0.40 originally; 0.20 in replications, with some effects reversed.

Open Science Collaboration, *Science*, 2015. Figure 1. doi:10.1126/science.aac4716

Planning and reporting

Before collecting data

Specify the primary outcome, hypotheses and significance level.
Choose a meaningful effect and a sample size that can detect it.

When interpreting results

Read the effect estimate and its confidence interval.
Which clinically important effects does the confidence interval rule out, and which does it still allow?

When reporting

Show the primary result and the relevant secondary results.
Account for multiple testing.

The aim is useful evidence for the question that justified the study.

References

1. Hayward G, Hay AD, Moore MV, et al. Effect of oral dexamethasone without immediate antibiotics vs placebo on acute sore throat in adults: a randomized clinical trial. *JAMA* 2017;317(15):1535. doi:[10.1001/jama.2017.3417](https://doi.org/10.1001/jama.2017.3417)
2. Munroe R. Significant. *xkcd* 882. xkcd.com/882/. Licensed CC BY-NC 2.5.
3. Kerr NL. HARKing: hypothesizing after the results are known. *Personality and Social Psychology Review* 1998;2(3):196. doi:[10.1207/s15327957pspr0203_4](https://doi.org/10.1207/s15327957pspr0203_4)
4. Open Science Collaboration. Estimating the reproducibility of psychological science. *Science* 2015;349(6251):aac4716. doi:[10.1126/science.aac4716](https://doi.org/10.1126/science.aac4716)

Thank you very much for your attention